

MATH 2400 - Week 3 Notes

1 Vector Functions

A **vector-valued function** or **vector function** is a function that maps real numbers to vectors of real numbers. We will work mostly in $\mathbb{R}^3$, where vector-valued functions take the form

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

Here, the **component functions** f, g, h are single-variable functions that you are familiar with from Calc 1 and Calc 2 (and everything else you've ever done with functions).

Limits and continuity for vector-valued functions are defined in terms of their components:

$$\lim_{t \rightarrow a} \vec{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle$$

and $\vec{r}(t)$ is continuous at a if

$$\lim_{t \rightarrow a} \vec{r}(t) = \vec{r}(a)$$

The graphs of vector-valued functions are 1-dimensional curves in 3-dimensional space, so they are called **space curves**. If we look at just the component functions, we get the **parametric equations** for the space curve C :

$$x = f(t), y = g(t), z = h(t)$$

Example: Find a vector function that represents the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $y + z = 2$.

In the xy -plane, the cylinder has parametric equations $x = \cos(t)$, $y = \sin(t)$, $0 \leq t \leq 2\pi$. Since the equation of the plane is $y + z = 2$, we have $z = 2 - y = 2 - \sin(t)$. This means that the parametric equations for C , the curve of intersection, can be written as $x = \cos(t)$, $y = \sin(t)$, $z = 2 - \sin(t)$, $0 \leq t \leq 2\pi$.

It is important to distinguish between particles that travel along paths (curves) that **intersect** and particles that **collide**. Particles that collide pass through the same point at the same time, whereas particles whose paths intersect (but don't collide) pass through the same point but at different times.

2 Derivatives of Vector-Valued Functions

Since derivatives are defined in terms of limits, we can define the **derivative** of $\vec{r}(t)$ in the usual way:

$$\vec{r}'(t) = \frac{d\vec{r}}{dt} = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}$$

Please note that the derivative of a vector-valued function is also a vector-valued function, so it is a vector, called the **tangent vector**. If $\vec{r}(t_0)$ has tail at the origin and head at the point P , then the tangent vector to the curve given by $\vec{r}(t)$ at the point P is the vector $\vec{r}'(t_0)$ with its tail at point P .

If $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, then the derivative is given by $\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$, where $f'(t), g'(t), h'(t)$ are the derivatives of $f(t), g(t), h(t)$.

2.1 Differentiation Rules

If $\vec{u}(t)$ and $\vec{v}(t)$ are differentiable vector-valued functions, c is a scalar, and $f(t)$ is a real-valued function, then:

1. $[\vec{u}(t) + \vec{v}(t)]' = \vec{u}'(t) + \vec{v}'(t)$
2. $[c\vec{u}(t)]' = c\vec{u}'(t)$
3. $[f(t)\vec{u}(t)]' = f'(t)\vec{u}(t) + f(t)\vec{u}'(t)$
4. $[\vec{u}(t) \cdot \vec{v}(t)]' = \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t)$
5. $[\vec{u}(t) \times \vec{v}(t)]' = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$
6. $[\vec{u}(f(t))]' = f'(t)\vec{u}'(f(t))$

From Rule 4 we get the following important result:

If $|\vec{r}(t)| = c$, where c is a constant, then $\vec{r}'(t) \cdot \vec{r}(t) = 0$ for all t (so the tangent vector is orthogonal to the position vector).

Start by writing $|\vec{r}(t)|^2 = \vec{r}(t) \cdot \vec{r}(t) = c^2$. Then, using Rule 4, we have that $[\vec{r}(t) \cdot \vec{r}(t)]' = 2\vec{r}'(t) \cdot \vec{r}(t) = 0$, which is what we wanted to show.

3 Integration of Vector-Valued Functions

Just like with derivatives, the definite integral of a vector-valued function, $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, is defined as the definite integral of its component functions:

$$\int_a^b \vec{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle$$

The Fundamental Theorem of Calculus extends in the natural way to vector-valued functions:

$$\int_a^b \vec{r}(t) dt = \vec{R}(b) - \vec{R}(a)$$

where $\vec{R}(t)$ is any antiderivative of $\vec{r}(t)$, that is, $\vec{R}'(t) = \vec{r}(t)$.

As with real-valued functions, we use indefinite integrals to denote the antiderivatives of vector-valued functions:

$$\int \vec{r}(t) dt = \vec{R}(t) + C$$

where C is the constant of integration.